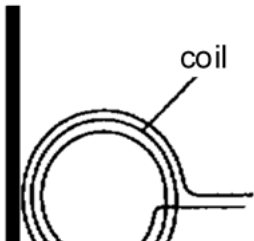


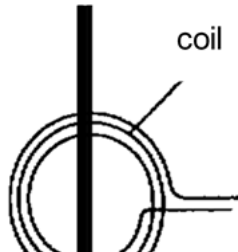
26 The figures below show the arrangement of a coil and a long straight cable placed near each other.

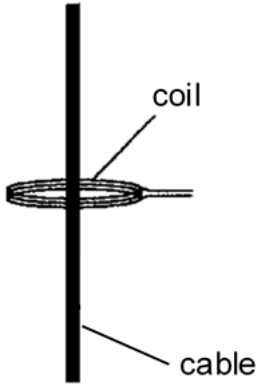
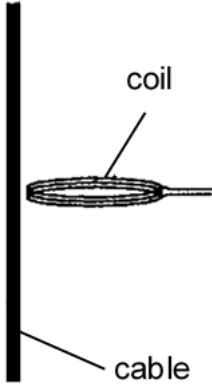
In which of the arrangement will the magnetic flux through the coil be the largest when a current flows through the cable?

A



B



	C		D	
	Answer: A Option B results in the magnetic flux to be zero as magnetic flux produced by the current in the wire flows in opposite directions on either side of the cable. Since magnetic flux density is a vector quantity, the vector sum will be zero through the coil. Options C and D are also zero as the cross sectional area of the coil is not perpendicular to the magnetic flux density that is flowing through it. Hence the product of the magnetic flux density and the cross sectional area is zero. That leaves only option A to have a net magnetic flux through the coil.			